Analyzing Cross-Lingual Topic Spaces and Story Representations in Pre-Trained Language Models

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1 Introduction

Storytelling is an inherently human creation brings communities together. Culture can be ingrained into these stories. Pre-trained models are good at taking in high resource languages and finding topics, this could be seen as a double sided sword when it comes to low/medium resourced languages. I tested whether three popular models can agree on topics from several low and medium resource languages. If we have models be trained on more languages there might be more representation on cultures which hopefully hopefully steer to less biased summarization/aid in better translation.

With these models, there will be biases from the languages with more data, so my main research question is How close are the topics of stories and spoken narratives from differing communities of low/medium resourced languages? I hypothesized that the topics will be similar by language families and will see pockets in the overall space.

2 Related work

Conneau et al. (2020) shows that pretraining multilingual language models will lead to higher performance gains for a wide range of cross-lingual transfer tasks. Conneau et al. (2020) develped the XLM-R which is the bases of the XLM-RoBerta (Conneau et al., 2019). Conneau et al. (2020) attempted to improve the cross-lingual language understanding (XLU) of the models. They did this by carefully studying the effects of the unspervised crosslingual training representations at a very large scale.

Eldan and Li (2023) shows that models around 125 million parameters can rarely generate coherent and consistent English text beyond a few words even after extensive training. They have produced the TinyStories dataset which is a synthetic dataset of short stories which uses words which is typical for a 3 to 4-year-olds. These stories were generated by GPT-3.5 and GPT-4. These stories can be used to train and evaluate models that are much smaller than the state-of-the-art models.

Adelani et al. (2022) has developed MasakhaNEWS which is the largest dataset for news topic classification. This dataset covers 16 languages widely spoken in Africa. This dataset has shaped how the development of several machine learning algorithms have been done, such as topic modeling.

Bianchi et al. (2021) has shown that many datasets co-exist in multiple languages. This covers the same content across the board. Bianchi et al. (2021) states that the linguistic differences in the datasets would make it impossible to use the more conventional and traditional, bag-of-word-based topic models. This is due to the struggles with cross-lingual applications due to the sparse vocabulary that some of the datasets provide with just monolingual corpora differences. Bianchi et al. (2021) states that models can take 2 form, either be monolingual or suffer from a sparse vocabulary wich can be troublesome when we want to do multilingual tasks. Transfer learning can play a role in mitigating this risk. Bianchi et al. (2021) introduces a zero-shot cross-lingual topic model that learns topics in English, and predicts them in documents in Italian, French, German, and Portuguese. This relates to this project by having a way to do multilingual zero-shot models. I am adding the option for low-resource languages and story telling to the mix.

3 Methodology

3.1 Pre-trained models

I have testing on three different models, two are from the Bert family: XLM-RoBERTa base (Conneau et al., 2019) and DeBerta v3 Base (He et al., 2021a,b) and a Qwen2.5-7B Instruct (Qwen Team, 2024; Yang et al., 2024) model. Since I am using

an Instruct model, one would think of adding instructions to get the desired output, I did not add any instructions and just pulled the pure embedding and raw outputs from this model.

3.2 Feasible tests

I have done two distinced umbrella tests using the models.

The first test is to visualize the embedding space of all of the languages. We can see the visualization in Figure 1 in Appendix A. With this embedding space, I calculated the cosine similarity for each model and the silhouette score per language. These tables can be found from Table 1 to 4 in Appendix A.

The second test, I introduced a Latent Dirichlet Allocation (LDA) (Blei et al., 2003) to produce some kind of gold standard. I am using the LDA to create a reference because the datasets were not created with a topic column so if I want to find any bias, I would need an unbiased creator. The topics are in the the actual language that the stories are in, trying to translate these topics into English would inroduce some influences and possibly remove any cultural neuance to the topics. The LDA model was prompted to create 5 categories for each language. Using these categories, fed the categories to the pre-trained models and found the distribution that the models had chose for the languages. This distribution can be found in Figure 2 in Appendix A. Since I have the distribution of the models, I calculated the Canonical Correlation Analysis (CCA) between the models for each language. The table can be found in Table 5 in Appendix A. Finally I calculated the Jensen-Shannon score of just the first two topic labels, which I will talk about in section 5.

4 Data

5 Results

6 Conclusion

Limitations

Acknowledgement

A Figures and Tables

Refer to Table 1 to 5 for the tables and Figure 1 and 2 for the two figures.

Language	DeBERTa	Qwen	XLM-RoBERTa
Acholi	0.1994	0.0989	0.0438
Luganda	0.1933	0.1253	0.0853
Masaaba	0.1800	0.0817	0.0814
Nyankole	0.1924	0.1210	0.0865
Soga	0.1911	0.1170	0.0946
Akan	0.1891	0.1277	0.0651
Ewe	0.1813	0.0633	0.0527
Dagbani	0.2281	0.0730	0.0509
Norwegian	0.1251	0.0663	0.0744
Slovak	0.1232	0.0656	0.0751
Serbian	0.1313	0.1568	0.1456
Bulgarian	0.1145	0.0710	0.0636
Czech	0.1216	0.0634	0.0741
Turkish	0.1389	0.0700	0.0819
Persian	0.1374	0.0818	0.0657
Vietnamese	0.1361	0.0614	0.0720
Indonesian	0.0958	0.0566	0.0727
Assamese	0.1815	0.0419	0.0648
Dogri	0.1576	0.0957	0.0506
GOM	0.1757	0.0730	0.0554
Kannada	0.1461	0.0429	0.1174
Nepali	0.1900	0.0551	0.0296
Tamil	0.1330	0.0468	0.0689
Sindhi	0.1466	0.0534	0.0461

Table 1: Silhouette Scores per Language Across Models

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9508
Acholi	Masaaba	0.9619
Acholi	Nyankole	0.9583
Acholi	Soga	0.9615
Acholi	Akan	0.9490
Acholi	Ewe	0.9513
Acholi	Dagbani	0.9727
Acholi	Norwegian	0.6627
Acholi	Slovak	0.6109
Acholi	Serbian	0.6337
Acholi	Bulgarian	0.5994
Acholi	Czech	0.5950
Acholi	Turkish	0.6592
Acholi	Persian	0.7178
Acholi	Vietnamese	0.4640
Acholi	Indonesian	0.6435
Acholi	Assamese	0.7712
Acholi	Dogri	0.8890
Acholi	GOM	0.9279
Acholi	Kannada	0.7280
Acholi	Nepali	0.7242
Acholi	Tamil	0.7609
Acholi	Sindhi	0.7811
Luganda	Nyankole	0.9677
Luganda	Soga	0.9679
Norwegian	Slovak	0.9842
Norwegian	Serbian	0.9835
Slovak	Czech	0.9935
Kannada	Nepali	0.9748
Nepali	Sindhi	0.9694

Table 2: Cross-Lingual Cosine Similarity Metrics (DeBERTa Model)

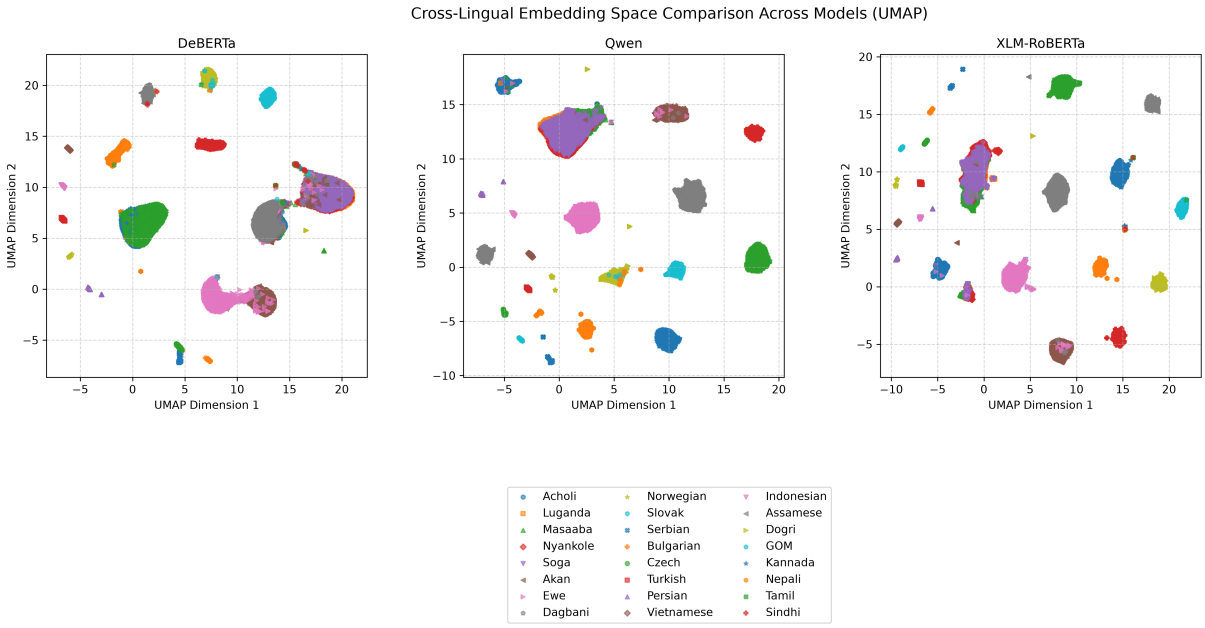


Figure 1: A minimal working example to demonstrate how to place two images side-by-side.

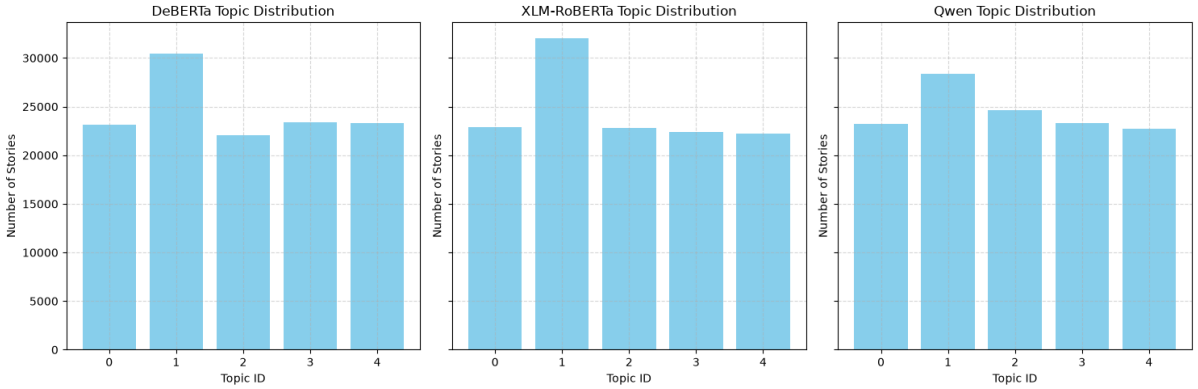


Figure 2: LDA topic categorization distribution

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9033
Acholi	Masaaba	0.8999
Acholi	Nyankole	0.9018
Acholi	Soga	0.9156
Acholi	Akan	0.8560
Acholi	Norwegian	0.5413
Acholi	Bulgarian	-0.1191
Acholi	Persian	-0.1207
Acholi	Kannada	-0.0689
Luganda	Nyankole	0.9606
Norwegian	Slovak	0.4964
Slovak	Czech	0.8998
Dogri	GOM	0.9543
Kannada	Tamil	0.5917

Table 3: Cross-Lingual Cosine Similarity Metrics (Qwen Model)

Language 1	Language 2	Mean Cosine Sim.
Acholi	Luganda	0.9965
Acholi	Masaaba	0.9962
Acholi	Nyankole	0.9967
Acholi	Norwegian	0.9894
Luganda	Nyankole	0.9971
Norwegian	Slovak	0.9974
Slovak	Czech	0.9987
Vietnamese	Indonesian	0.9983
Dogri	GOM	0.9980
Kannada	Tamil	0.9975

Table 4: Cross-Lingual Cosine Similarity Metrics (XLM-RoBERTa Model)

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	Language	Comparison	Dimension 1	Dimension 2	Dimension 3
0	Acholi	DeBERTa vs Qwen	0.999996	0.994984	0.996444
1	Acholi	DeBERTa vs XLM-RoBERTa	0.996180	0.983579	0.979210
2	Luganda	DeBERTa vs Qwen	0.997958	0.996214	0.992004
3	Luganda	DeBERTa vs XLM-RoBERTa	0.992680	0.980998	0.978608
4	Masaaba	DeBERTa vs Qwen	0.996729	0.994966	0.990517
5	Masaaba	DeBERTa vs XLM-RoBERTa	0.993225	0.980985	0.979801
6	Nyankole	DeBERTa vs Qwen	0.994315	0.991878	0.984842
7	Nyankole	DeBERTa vs XLM-RoBERTa	0.988397	0.982236	0.973999
8	Soga	DeBERTa vs Qwen	0.996412	0.994938	0.989129
9	Soga	DeBERTa vs XLM-RoBERTa	0.992065	0.984717	0.972391
10	Akan	DeBERTa vs Qwen	0.988052	0.992737	0.988121
11	Akan	DeBERTa vs XLM-RoBERTa	0.988035	0.983507	0.979119
12	Ewe	DeBERTa vs Qwen	0.992485	0.988232	0.986231
13	Ewe	DeBERTa vs XLM-RoBERTa	0.985299	0.987757	0.976501
14	Dagbani	DeBERTa vs Qwen	0.992607	0.984602	0.984641
15	Dagbani	DeBERTa vs XLM-RoBERTa	0.990721	0.978886	0.982787
16	Norwegian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
17	Norwegian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
18	Slovak	DeBERTa vs Qwen	1.000000	1.000000	1.000000
19	Slovak	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
20	Serbian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
21	Serbian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
22	Bulgarian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
23	Bulgarian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
24	Czech	DeBERTa vs Qwen	1.000000	1.000000	1.000000
25	Czech	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
26	Turkish	DeBERTa vs Qwen	1.000000	1.000000	1.000000
27	Turkish	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
28	Persian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
29	Persian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
30	Vietnamese	DeBERTa vs Qwen	1.000000	1.000000	1.000000
31	Vietnamese	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
32	Indonesian	DeBERTa vs Qwen	1.000000	1.000000	1.000000
33	Indonesian	DeBERTa vs XLM-RoBERTa	1.000000	1.000000	1.000000
34	Assamese	DeBERTa vs Qwen	0.998580	0.998028	0.997125
35	Assamese	DeBERTa vs XLM-RoBERTa	0.992181	0.993391	0.987680
36	Dogri	DeBERTa vs Qwen	0.998329	0.998787	0.997214
37	Dogri	DeBERTa vs XLM-RoBERTa	0.992667	0.989423	0.984441
38	GOM	DeBERTa vs Qwen	0.998587	0.998678	0.997664
39	GOM	DeBERTa vs XLM-RoBERTa	0.993198	0.991190	0.983822
40	Kannada	DeBERTa vs Qwen	0.994520	0.976164	0.971726
41	Kannada	DeBERTa vs XLM-RoBERTa	0.992557	0.988733	0.984590
42	Nepali	DeBERTa vs Qwen	0.998997	0.998740	0.998223
43	Nepali	DeBERTa vs XLM-RoBERTa	0.991894	0.992935	0.986940
44	Tamil	DeBERTa vs Qwen	0.984171	0.978970	0.974888
45	Tamil	DeBERTa vs XLM-RoBERTa	0.991453	0.987195	0.984323
46	Sindhi	DeBERTa vs Qwen	0.998605	0.999204	0.998230
47	Sindhi	DeBERTa vs XLM-RoBERTa	0.992767	0.991836	0.985045

Table 5: CCA for each language against each model